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SEMINAR AND PROJECT INITIATION COURSE

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Topic : IoT in Smart Homes

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Abstract

- IoT connects home sensors, appliances and controllers so they can sense, communicate and act automatically
- Seminar covers architecture, devices, protocols and applications
- Main open problems: security, privacy and interoperability between vendors
- Matter standard unifies devices over Wi-Fi and Thread
- Ends with a project initiation plan for a local, low-cost smart home

- **Keywords: IoT, smart home, Matter, Zigbee, security**

1. Introduction

- Smart home: lights, locks, cameras, thermostats and appliances connected to a network
- IoT: small devices with sensors, a processor and a network link that exchange data
- **How it works**
  - Sensors collect data (temperature, motion, door, power)
  - Hub or router carries it over Wi-Fi, Bluetooth, Zigbee, Z-Wave or Thread
  - Software applies rules in the cloud or locally
  - Actuators act: lights, locks, thermostat; user controls it from an app
- Why it matters: homes are where most people meet IoT, early devices had weak security, brands often do not interoperate

2. Literature Survey (1/4)

Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
1	Introductory IoT guide (Ananna & Saifuzzaman, 2023)	Sensors, network, cloud and apps (general IoT architecture)	None; a survey-style guide, no experimental dataset	To explain IoT from scratch for beginners.	Names security and privacy as the big open problems; no system is built or tested.
2	Low-cost smart home automation (Stolojescu-Crisan et al., 2021)	Bluetooth, Zigbee and Wi-Fi compared; phone app	Sensor and control data from a small home prototype	To compare Bluetooth, Zigbee and Wi-Fi and build a low-cost smart home controlled from a phone app.	Small home prototype; security and interoperability between brands are not the focus.
3	Smart home monitoring with Android app (Akter et al., 2023)	Sensors, microcontroller, Wi-Fi, Android app	Sensor readings from the prototype	To monitor and automate a home, showing status and alerts in a mobile app.	Depends on Wi-Fi and a phone app; no local dashboard or offline operation.

2. Literature Survey (2/4)

Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
4	Wi-Fi based home automation (Sachdeva, 2018)	NodeMCU, relays, Wi-Fi (Bluetooth and Zigbee compared)	Switching and control data from the prototype	To switch home devices cheaply through relays using NodeMCU and Wi-Fi.	Basic on/off control over Wi-Fi; limited security and automation rules.
5	Comparative literature survey of 20 IoT smart home works (Kaur et al., n.d.)	Arduino, Raspberry Pi, ESP8266, ZigBee, Wi-Fi, GSM; web and mobile apps (Blynk)	None; compares earlier studies in two tables	To compare IoT smart home systems by issue addressed, merit, demerit, communication, controller, storage and tools.	Review only; no system is built; many compared works list no demerits; mostly Wi-Fi and Arduino based.
6	Voice and web controlled home automation (Nadgaundi et al., 2020)	PIC16F877A, GSM SIM900A, Bluetooth HC-05, PHP and MySQL web page, SS Voice Android app	None; sensor status (LPG, PIR, IR, water level) logged online	To build a low-cost system with voice and website control plus gas, burglar and water-level safety, helpful for elderly and disabled users.	Semi-automated, needs a user command; Bluetooth works only in offline mode; only eight fixed voice commands.

2. Literature Survey (3/4)

Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
7	Smart home automation system based on IoT (Kadiyan et al., 2023)	ESP32 NodeMCU, four relays, Blynk app, Wi-Fi	None; proposed architecture and block diagram	To propose a scalable architecture for remote control, monitoring and rule-based automation of lights and a fan.	Largely conceptual; no test results; security (TLS, OAuth) is described but not implemented.
8	IoT smart fire and gas safety system for rural homes (Sharma et al., 2026)	ESP32, MQ2, MQ5, DHT11, buzzer, Blynk	Readings from a controlled kitchen test; no public dataset	To give low-cost real-time smoke and LPG leak alerts; buzzer in about 2.4 s, phone alert in about 3.5 s, 0.5 to 1 W power.	Tested only in a controlled kitchen; MQ sensors are cross-sensitive and need warm-up; DHT11 is low precision; remote alerts need Wi-Fi.
9	Impact of IoT on smart home energy management (Kumar & Pandey, 2023)	Smart meters, thermostats, smart appliances; Python analysis; Homergy box (Arduino Mega, NodeMCU, relays)	Assumed CSV of home energy use before and after IoT	To study how IoT monitoring and automation reduce energy use and cost; reports 500 to 400 kWh per month (20%).	Dataset is assumed, so results are illustrative; privacy, security and interoperability remain open.

2. Literature Survey (4/4)

Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
10	Home automation using ThingSpeak (Akhtar et al., 2019)	Arduino UNO, Raspberry Pi 3, ThingSpeak cloud, Android app, PIR, LDR, soil moisture sensor, servo lock	ThingSpeak logs and monthly kWh readings, January to May 2019, with and without the system	To control lights, fans, sprinkler and door lock remotely and monitor energy use; energy use was lower with the system.	Needs continuous power and internet; no battery backup; depends on the ThingSpeak cloud.
11	Smart Home Switch System using IoT (Ab Rahim & Yahaya, 2023)	ESP32, ACS712 current sensor, DHT11, 8-channel relay, Blynk	None; live current, power, temperature and humidity readings	To control four loads from a phone and monitor current, power and room conditions.	Power is estimated assuming a constant 240 V supply; DHT11 is low precision; cloud and Wi-Fi dependent; no quantified results.
12	Blynk framework for smart home (Media's et al., 2019)	Raspberry Pi as server, NodeMCU clients, DHT11, relay, Blynk	None; prototype with event logs stored on the Raspberry Pi	To retrofit an existing house cheaply with wireless nodes and no rewiring, controlled by a Blynk phone app.	Very small prototype (one sensor, one relay); NodeMCU I/O load is limited; needs backup power; security not studied.

3. Information: architecture, devices

- **Five-layer architecture**
  - Perception: sensors and actuators
  - Network: Wi-Fi, Bluetooth, Zigbee, Z-Wave, Thread
  - Middleware: gateway or cloud, rules, device management
  - Application: mobile app, voice assistant, dashboard
  - Business: energy billing, maintenance services
- **Devices**
  - Sensors: temperature, motion, door, smoke, water leak
  - Actuators: bulbs, plugs, locks, thermostats; Controllers: hub, speaker, Raspberry Pi

3. Information: applications, security

- **Applications**
  - Lighting and comfort: schedules, motion lights, voice control
  - Security: locks, cameras, door sensors, phone alerts
  - Energy: smart plugs, thermostats (10 to 23 percent saving reported for learning thermostats)
  - Health: fall detection and activity monitoring for the elderly
- **Security and privacy risks**
  - Buffer overflows (28.25%) and DoS flaws (27.20%) are the most common vulnerabilities
  - Default passwords still enable botnets such as Mirai
  - 76% of users know the risks; only 24% update firmware regularly
  - Defence: change passwords, update firmware, separate IoT network, local processing

4. Advantages / Disadvantages

Advantages	Disadvantages
Convenience: one app or voice control	Security risks: default passwords, unpatched firmware
Energy saving through schedules and thermostats	Privacy: usage data reveals when people are home
Safety: smoke, gas and leak alerts	Interoperability gaps between brands
Helps elderly and disabled people	Device and hub cost; vendor support ends
Automation without effort	Depends on internet and power; cloud outages

5. Conclusion

- IoT makes homes more convenient, safer and more energy efficient
- Surveys agree: security, privacy and interoperability are the main weaknesses
- Matter, regular updates and local processing address them
- A simple local low-cost system is a practical next step (Section 6)

6. Project Initiation

- **Proposed project: local, low-cost smart home controller**
  - ESP32 sensor and relay nodes report over Wi-Fi using MQTT
  - One dashboard on a Raspberry Pi; works without the cloud
  - Covers: problem statement, scope and objectives, stakeholders, execution plan, resources

6.1 Description of problem statement

- Most commercial smart home devices depend on vendor clouds
- Default settings are weak and different brands do not work together
- Home users and students lack a simple, low-cost, private system that is easy to extend
- **Problem: build a local smart home with sensor and relay nodes and one dashboard, without cloud dependency**

6.2 Scope and objectives

- **Scope**
  - One room or flat; sensing (temperature, humidity, motion, door); control (lights, fan); dashboard and alerts
  - Out of scope: cameras, voice assistants, cloud services
- **Objectives**
  - Study IoT architecture, protocols and security risks
  - Build sensor and relay nodes reporting to a local controller
  - Provide a dashboard and simple automation rules
  - Apply basic security: new passwords, separate network, updates
  - Test reliability and response time; document results

6.3 Stakeholders

Stakeholder	Role	Interest
Student	Designer and builder	Complete seminar and project
Guide (faculty)	Supervisor	Technical quality, timely progress
Department / HOD	Approver	Course outcomes of SPI - 315003
Home users (family)	End users	Easy control, safety, privacy

6.4 Seminar execution plan

Week	Activity
1	Topic selection and literature survey
2	Gap, problem statement and objectives
3	Prepare report and presentation; guide review
4	Deliver seminar and collect feedback

6.5 Platform/ Equipment/ Resources

What we need to buy

- ESP32 board (a small Wi-Fi brain)
- Temperature sensor
- Motion sensor
- Door sensor
- Relay switch (turns a light or fan on and off)

Main controller

- Raspberry Pi (a tiny computer) or a laptop
- Connected to the home Wi-Fi
- Keeps everything running at home

Free software

- Mosquitto: passes messages between devices
- Home Assistant or Node-RED: makes the dashboard
- Arduino IDE: to write code for the boards

Small tools

- Wires to join the parts
- A plug-in test board for trying the wiring
- A small meter to check power
- A phone to view the dashboard

Bibliography

- **Papers**
  - Ananna & Saifuzzaman, Introduction to IoT (2023)
  - Stolojescu-Crisan, Crisan, Butunoi, An IoT-Based Smart Home Automation System (2021)
  - Akter et al., IoT-based Smart Home Monitoring and Automation System Using Android Application (2023)
  - Sachdeva, Design and Implementation of Wi-Fi based Smart Home Automation System, M.Tech thesis (2018)
- **Websites**
  - Silicon Labs: Matter 1.4 Continues the CSA's Commitment to Unifying the Home
  - The Matter Standard in 2026: A Status Review (matter-smarthome.de)

References

[1] E. N. Ganesh et al., "Implementation of IOT Architecture for SMART HOME using GSM Technology," International Journal of Computer Techniques, vol. 4, issue 1, Jan–Feb 2017.

[2] Laila Salman et al., "Energy efficient IoT-based smart home," in Proc. IEEE 3rd World Forum on Internet of Things (WF-IoT), 2016.

[3] Nasir Nazir Khan et al., "IoT-Based Smart Home Implementation Using Cayenne Web Server and Android SDK," International Journal of Advanced Science and Technology, vol. 29, no. 4, pp. 4034–4041, 2020.

[4] Sung G-M et al., "Internet of Things–based smart home system using a virtualized cloud server and mobile phone app," International Journal of Distributed Sensor Networks, September 2019.

Thank you